



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12

MATHEMATICS EXTENSION 1

HALF-YEARLY EXAMINATION

ASSESSMENT TASK 3

Time Allowed - two hours
(Plus 5 minutes reading time)

Weighting - 30%

Morning Session

Friday 1st May 2009

General Instructions

- Write using blue or black pen.
- Begin each question on a new page.
- Place your number, class and teacher at the top of each page.
- Attempt ALL questions 1 - 7.
- All questions are of equal value.
- Mark values are shown at the side of each question.
- All necessary working should be shown in every question.
- Board-approved calculators may be used.
- Hand in a page for each question even if the question was not attempted.

Question 1. (12 Marks)**(Start this question on a NEW page)****Marks**

- (a) Find, correct to the nearest minute, the acute angle between the lines
 $2y - x - 1 = 0$ and $y - 3x + 2 = 0$

2

- (b) Find the exact value of $\sin \frac{\pi}{12}$.

2

- (c) Find the coordinates of a point P which divides the interval AB internally in the ratio of 1:3, if A is (-5, -1) and B is (3, 7).

2

- (d) Solve $\frac{1}{x(x-1)} \leq 0$ and graph the solution on a number line.

3

- (e) State the domain and range of $f(x) = 3 \cos^{-1} 2x$ and sketch the curve.

3

Question 2. (12 Marks)

(Start this question on a NEW page)

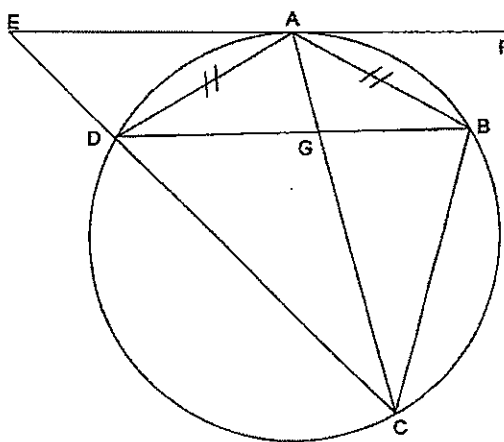
Marks

- (a) Find the general solution of the equation $2\sin\theta - \sqrt{3} = 0$.

2

(b)

ABCD is a quadrilateral inscribed in a circle with $AB = AD$. The tangent at A meets CD produced at E.



Copy the diagram on to your paper.

- (i) Prove $EF \parallel DB$.

2

- (ii) Prove AC bisects $\angle BCD$.

2

- (iii) Prove that $\triangle EAD \sim \triangle ACB$.

2

- (c) Find the volume of the solid generated when the region between the curve $y = x^2 - 4$ and the line $y = 2x - 4$ is rotated about the y -axis.

4

Question 3. (12 Marks)

(Start this question on a NEW page)

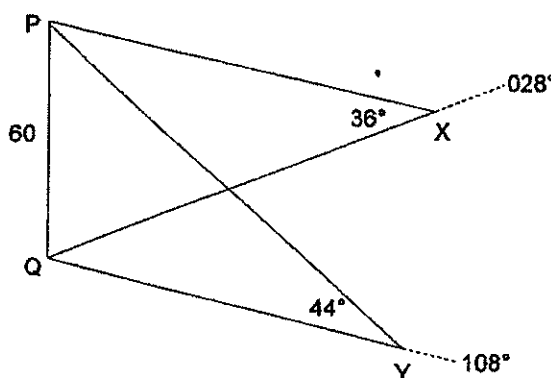
Marks

(a) Use the substitution $u = \sin 2x$ to evaluate $\int_0^{\frac{\pi}{4}} \sin^4 2x \cos 2x \, dx$ 3

(b) Using the principles of mathematical induction show that $\sum_{r=1}^n (3r-2) = \frac{n(3n-1)}{2}$ for all positive integers n . 3

(c) Solve $(w^2 - 2w)^2 - 2(w^2 - 2w) - 3 = 0$. 3

- (d) From an observation tower PQ of height 60 metres, two points X and Y at ground level have bearings 028° and 108° . From the top of the tower, the angles of depression of X and Y are 36° and 44° respectively.



(i) Prove that $XY^2 = 60^2 (\cot^2 36^\circ + \cot^2 44^\circ - 2 \cot 36^\circ \cot 44^\circ \cos 80^\circ)$ 2

(ii) Find XY to the nearest metre. 1

Question 4. (12 Marks) (Start this question on a NEW page)**Marks**

- (a) A spherical bubble is expanding so that its volume is increasing at the constant rate $10 \text{ cm}^3/\text{sec}$. What is the rate of increase of the radius when the surface area is $500 \text{ cm}^2/\text{sec}$?

2

(b) (i) Find $\int \frac{1}{\sqrt{9-4x^2}} dx$

2

(ii) Evaluate $\int_0^{\pi} \frac{1}{2x+\pi} dx$

2

- (c) Given the function $y = \log_e \sqrt{x^2 - 1}$, find:

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

3

- (d) Find the equation of the tangent to the curve $y = \sin^{-1} \frac{x}{2}$ at the point where $x = \sqrt{2}$.

3

Question 5. (12 Marks)**(Start this question on a NEW page)****Marks**

- (a) (i) Express $\sqrt{3} \cos x - \sin x$ in the form of $R \cos(x + \alpha)$ 2
- (ii) Hence or otherwise solve $\sqrt{3} \cos x - \sin x = 1$ where $0 \leq x \leq 2\pi$. 2
- (b) Consider the function $f(x) = e^x - 1 - x$.
- (i) Show that the minimum of $f(x)$ occurs at $x = 0$. 2
- (ii) Deduce that $f(x) \geq 0$ for all x . 1
- (iii) On the same set of axes, sketch $y = e^x - 1$ and $y = x$. 2
- (iv) Find the inverse of $g(x) = e^x - 1$. 1
- (v) State the domain of $g^{-1}(x)$. 1
- (vi) For what values of x is $\log_e(1+x) \leq x$ 1

Question 6. (12 Marks)

(Start this question on a NEW page)

Marks

- (a) In a game of Sic Bo, three regular six-sided dice are thrown once.
Find the probability:

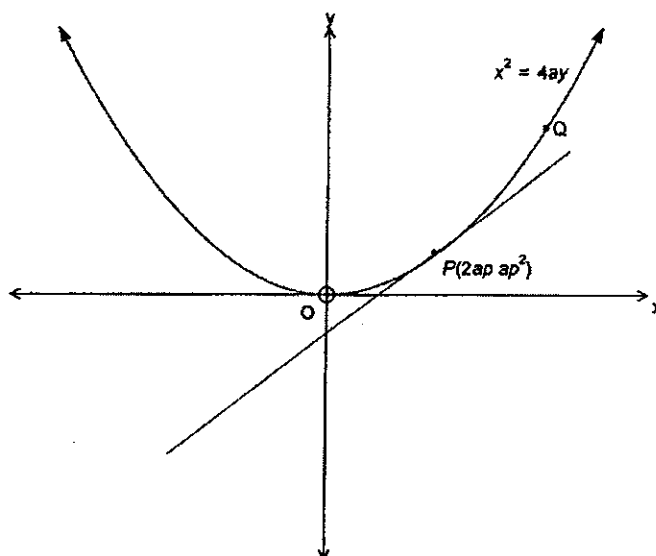
(i) of throwing three different numbers on the dice.

2

(ii) that exactly two of the dice show the same number.

2

- (b) The variable point $P(2ap, ap^2)$ lies on $x^2 = 4ay$, and the chord OQ is drawn parallel to the tangent at P . The tangents at P and Q meet at R .



- (i) Write down the equation of the chord OQ .
- (ii) Show that the coordinates of Q are $(4ap, 4ap^2)$.
- (iii) Find the equation of the tangent at Q .
- (iv) Show that R has the coordinates $(3ap, 2ap^2)$.
- (v) Find the Cartesian equation of the locus of R as P varies on the parabola.

1

2

1

2

2

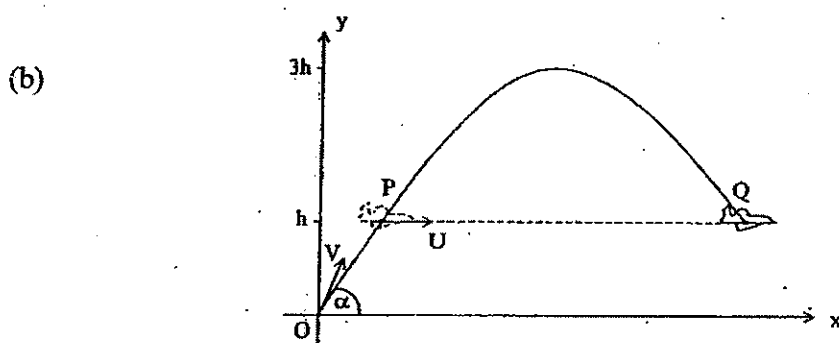
Question 7. (12 Marks)**(Start this question on a NEW page)****Marks**

- (a) The velocity $v \text{ ms}^{-1}$ of a particle moving in a straight line is given by $v = 6 - 2x$, where x is the displacement in metres from the origin O and t is the time in seconds. Initially the particle is at rest at the origin.

(i) Find the acceleration at the origin. 2

(ii) Show that $t = -\frac{1}{2} \log_e \left(1 - \frac{1}{3}x \right)$ 2

(iii) Find x as a function of t . 2



An enemy fighter plane is flying horizontally at height h metres with a speed $U \text{ ms}^{-1}$. When it is at point P a ground rocket is fired towards it from the origin O with a speed $V \text{ ms}^{-1}$ and angle of elevation α .

The rocket misses the plane, passing too late through point P . However, it goes on to reach a maximum height of $3h$ metres and then on its descent strikes the plane at Q . With the axes shown in the diagram above, you may assume that the position of the rocket is given by $x = Vt \cos \alpha$ and $y = -\frac{1}{2}gt^2 + Vt \sin \alpha$ where t is the time in seconds after firing and g is the acceleration due to gravity.

- (i) Show that initially the vertical component of the rocket's speed is $V \sin \alpha = \sqrt{6gh}$. 2
- (ii) If the rocket had not struck the fighter plane at Q , it would have returned to the x -axis at a distance d from O . Show that the horizontal component of the speed of the rocket is $V \cos \alpha = \frac{gd}{2\sqrt{6gh}}$. 2
- (iii) Show that the equation of the path of the rocket is $y = \frac{12hx}{d} \left(1 - \frac{x}{d} \right)$. 2

TGS - Year 12 - Extension 1 Mathematics - May 2009

$$1. a) \tan \theta = \frac{3 - \frac{1}{2}}{1 + 3(\frac{1}{2})} \quad (M1)$$

$$\theta = \tan^{-1}(1)$$

$$\theta = 45^\circ \quad (1)$$

$$(b) \frac{1}{x(x-1)} \leq 0 \quad x \neq 0, x \neq 1$$

$$x(x-1) \leq 0 \quad (M1)$$

$$\therefore 0 < x < 1$$



$$(c) P \left[\frac{1(3) + 3(-5)}{1+3}, \frac{1(7) + 3(-1)}{1+3} \right] \quad (M1)$$

$$= (-3, 1) \quad (1)$$

$$(d) \sin \frac{\pi}{12} = \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \quad (M1)$$

$$= \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} \quad (M1)$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \quad (1)$$

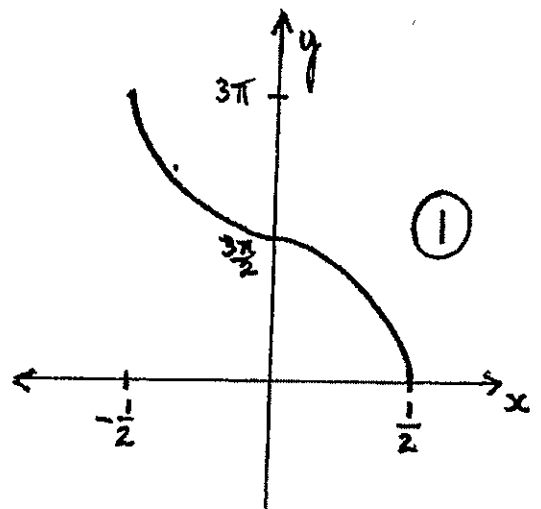
$$= \frac{\sqrt{2}(\sqrt{3} - 1)}{4}$$

$$(e) D: -1 \leq 2x \leq 1$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2} \quad (1)$$

$$R: 0 \leq \cos^{-1} 2x \leq \pi$$

$$0 \leq 3 \cos^{-1} 2x \leq 3\pi \quad (1)$$



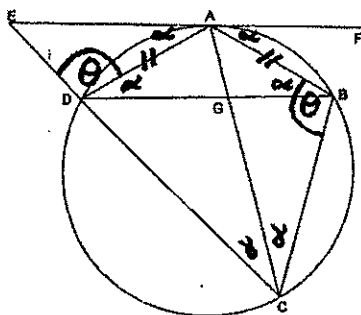
$$2. (a) \quad 2 \sin \theta - \sqrt{3} = 0$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{3} \quad (1)$$

$$\theta = n\pi + (-1)^n \left(\frac{\pi}{3}\right) \quad (1)$$

(b)



i) $\angle ADG = \angle ABG$ Base $\angle$'s isos. $\Delta =$
 $\angle FAB = \angle ADG$ $\angle$ between tang. & chord = $\angle$ in alt. seg.
 $\therefore \angle FAB = \angle ABG$ Equal $\angle$'s (1)
 $EF \parallel DB$ Alt. $\angle$'s =. (1)

(ii) $\angle EAD = \angle ABG$ $\left\{ \begin{array}{l} \angle \text{ between tang. \& chord} = \angle \text{ in alt. seg.} \\ \angle EAD = \angle ACD \end{array} \right.$ (1)
 $\angle FAB = \angle BCA$ (1)
 $\therefore \angle ACD = \angle BCA$ Equal angles (1)

iii) $\angle EAD = \angle BCA$ Equal angles
 $\angle EDA = \angle ABC$ Ext. $\angle$ cyc. quad = int. opp. $\angle$. (1)
 $\therefore \triangle EAD \sim \triangle ACB$ Two $\angle$'s $\Delta =$. (1)

$$c) \quad y = x^2 - 4 \quad y = 2x - 4$$

$$x^2 - 4 = 2x - 4$$

$$x(x-2) = 0$$

$$x = 0 \quad x = 2$$

$$y = -4 \quad y = 0$$

(1)

$$V = \pi \int_{-4}^0 (y+4) dy - \pi \int_{-4}^0 \left(\frac{y+4}{2}\right)^2 dy \quad (1)$$

$$= \pi \int_{-4}^0 \left[(y+4) - \left(\frac{1}{4}y^2 + 2y + 4\right) \right] dy$$

$$= \pi \int_{-4}^0 \left(-y - \frac{1}{4}y^2 \right) dy$$

$$= \pi \int_{-4}^0 \left(y + \frac{1}{4}y^2 \right) dy \quad (M1)$$

$$= \pi \left[\frac{y^2}{2} + \frac{y^3}{12} \right]_0^{-4}$$

$$= \pi \left(8 - \frac{16}{3} \right)$$

$$= \frac{8\pi}{3} \text{ units}^3 \quad (1)$$

$$\begin{aligned}
 3. a) \quad & \int_0^{\frac{\pi}{4}} \sin^4 2x \cos 2x \, dx \\
 &= \frac{1}{2} \int_0^1 u^4 \, du \\
 &= \frac{1}{2} \left[\frac{u^5}{5} \right]_0^1 \quad (M1) \\
 &= \frac{1}{10} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 u &= \sin 2x \\
 du &= 2 \cos 2x \, dx \\
 \text{when } x=0 \quad u &= 0 \\
 x=\frac{\pi}{4} \quad u &= 1 \quad (M1)
 \end{aligned}$$

$$(b) \quad 1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

$$\text{let } n=1$$

$$L.H.S = 1$$

$$RHS = \frac{1(3-1)}{2} = 1$$

True for $n=1$

Assume true for $n=k$.

$$S_k \rightarrow 1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$$

Prove true for $n=k+1$

$$[1 + 4 + 7 + \dots + 3k-2] + [3(k+1)-2]$$

$$= \frac{k(3k-1)}{2} + 3k+1$$

$$= \frac{k(3k-1) + 6k+2}{2} \quad (M1)$$

$$= \frac{3k^2 + 5k + 2}{2}$$

$$= \frac{(3k+2)(k+1)}{2} = R.H.S. \quad (1)$$

If true for $n=k$, we have proven true for $n=k+1$; since true for $n=1$, then true for $n=1+1$, true for $n=2, 3, 4, \dots$ for all positive integers n . (1)

$$(c) \quad \text{let } t = w^2 - 2w$$

$$t^2 - 2t - 3 = 0 \quad (M1)$$

$$(t-3)(t+1) = 0$$

$$(w^2 - 2w - 3)(w^2 - 2w + 1) = 0 \quad (M1)$$

$$(w-3)(w+1)(w-1)^2 = 0 \quad (1)$$

$$\therefore w = \pm 1, 3$$

$$(d) i) \quad QY = 60 \cot 44^\circ$$

$$QX = 60 \cot 36^\circ$$

$$\angle XQY = 108^\circ - 28^\circ = 80^\circ \quad (M1)$$

In ΔXQY

$$\begin{aligned}
 XY^2 &= (60 \cot 44^\circ)^2 + (60 \cot 36^\circ)^2 - 2(60 \cot 36^\circ)(60 \cot 44^\circ)(\cos 80^\circ) \quad (M1) \\
 &= 60^2 [\cot^2 44^\circ + \cot^2 36^\circ - 2 \cot 36^\circ \cot 44^\circ \cos 80^\circ]
 \end{aligned}$$

$$ii) \quad XY = 94.33$$

$$= 94 \text{ m} \quad (1) \quad [\text{must be correct to nearest metre}]$$

$$4. a) V = \frac{4}{3} \pi r^3$$

$$S = 4 \pi r^2$$

$$500 = 4 \pi r^2$$

$$\frac{dV}{dt} = 10$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{10}{4 \pi r^2} \quad (M1)$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{10}{500} \\ &= \frac{1}{50} \text{ cm/sec.} \quad (1) \end{aligned}$$

$$\begin{aligned} b)(i) \int \frac{1}{\sqrt{9-4x^2}} dx \\ &= \frac{1}{2} \int \frac{1}{\sqrt{\frac{9}{4}-x^2}} dx \quad (1) \\ &= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C \quad (1) \end{aligned}$$

$$\begin{aligned} ii) \int_0^{\pi} \frac{1}{2x+\pi} dx \\ &= \frac{1}{2} \left[\ln(2x+\pi) \right]_0^{\pi} \quad (1) \\ &= \frac{1}{2} \left[\ln 3\pi - \ln \pi \right] \\ &= \frac{1}{2} \ln 3 \quad (1) \end{aligned}$$

$$(c) \quad y = \frac{1}{2} \ln(x^2-1)$$

$$\begin{aligned} (i) \quad \frac{dy}{dx} &= \frac{1}{2} \frac{2x}{(x^2-1)} \\ &= \frac{x}{x^2-1} \quad (1) \end{aligned}$$

$$\begin{aligned} ii) \quad \frac{d^2y}{dx^2} &= \frac{(x^2-1)(1) - (x)(2x)}{(x^2-1)^2} \quad (M1) \\ &= \frac{-x^2-1}{(x^2-1)^2} \quad (1) \end{aligned}$$

$$\begin{aligned} (d) \quad y &= \sin^{-1}\left(\frac{x}{2}\right) \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1-\frac{x^2}{4}}} \cdot \frac{1}{2} \\ &= \frac{1}{\sqrt{4-x^2}} \\ m_T &= \frac{1}{\sqrt{2}} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{when } x &= \sqrt{2} \\ y &= \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) \\ &= \frac{\pi}{4} \\ (M1) \quad (\sqrt{2}, \frac{\pi}{4}) \\ y - \frac{\pi}{4} &= \frac{1}{\sqrt{2}}(x - \sqrt{2}) \\ y &= \frac{1}{\sqrt{2}}x + \frac{\pi}{4} - 1 \quad (1) \end{aligned}$$

$$5.(a)(i) \sqrt{3} \cos x - \sin x = R [\cos x \cos \alpha - \sin x \sin \alpha]$$

$$R \cos \alpha = \sqrt{3} \quad R \sin \alpha = 1$$

$$R^2 = 3+1 \quad \tan \alpha = \frac{1}{\sqrt{3}} \quad (M1)$$

$$\therefore R = 2 \quad \therefore \alpha = \frac{\pi}{6}$$

$$\sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6}) \quad (1)$$

$$(ii) 2 \cos(x + \frac{\pi}{6}) = 1$$

$$\cos(x + \frac{\pi}{6}) = \frac{1}{2} \quad (M1)$$

$$x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = \frac{\pi}{6}, \frac{3\pi}{2} \quad (1)$$

$$(b)(i) f(x) = e^x - 1 - x$$

$$f'(x) = e^x - 1$$

$$f''(x) = e^x$$

$$f(0) = 1 - 1 - 0 = 0 \quad (0,0)$$

Stat. Points
Max/Min occur when $f'(x) = 0$

$$e^x - 1 = 0 \quad (1)$$

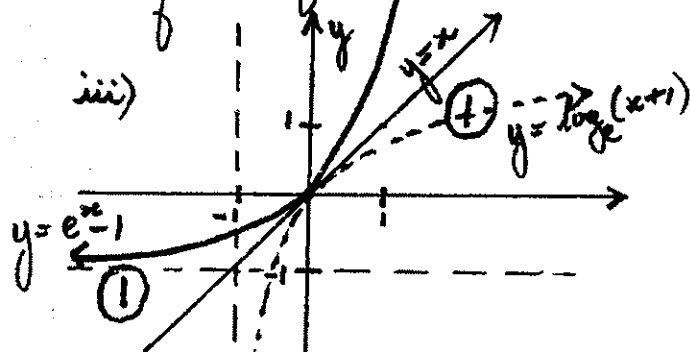
$$e^x = 1$$

$$\therefore x = 0$$

$$f''(0) = 1 \quad f''(0) > 0 \quad (1)$$

Minimum.

(ii) Stat. pt. occurs at $(0,0)$ & the minimum value of $f(x) = 0$ when $x = 0$. $\therefore f(x) \geq 0 \quad (1)$



iv) $g(x) = y = e^x - 1$

$$x = e^y - 1$$

$$x + 1 = e^y$$

$$g'(x) = y = \log_e(x+1) \quad (1)$$

v) D: $x + 1 > 0$

$$x > -1 \quad (1)$$

(vi) Check the graphs of $y = \log_e(x+1)$ and $y = x$

$$\log_e(x+1) \leq x \quad \text{for all } x > -1 \quad (1)$$

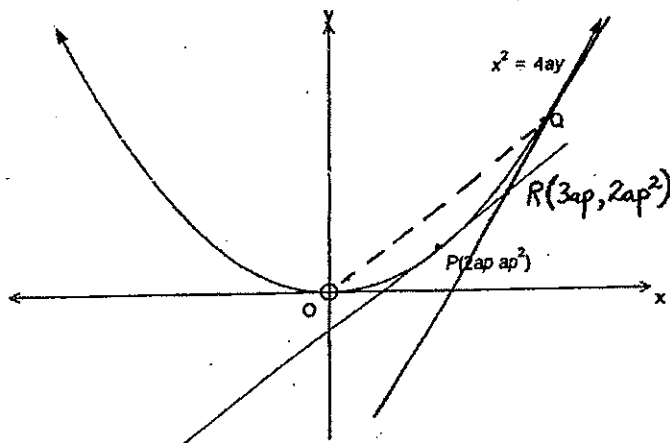
$$6.(a) \text{ i) } P(3 \text{ different nos.}) = 1 \times \frac{5}{6} \times \frac{4}{6} \quad (M1)$$

$$= \frac{5}{9} \quad (1)$$

$$\text{ii) } P(2 \text{ dice have same number}) = \left[\frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \right] \times 3 \times 6 \quad (M1)$$

$$= \frac{5}{12} \quad (1)$$

b)



$$x^2 = 4ay$$

$$y = \left(\frac{1}{4a}\right)x^2$$

$$y' = \left(\frac{1}{2a}\right)x$$

$$(i) \quad m_{\text{tangent at P}} = p$$

$$\text{equation of chord } y = px \quad (1)$$

$$\text{ii) } px = y \text{ and } x^2 = 4ay$$

$$\therefore x^2 = 4apx$$

$$x(x - 4ap) = 0 \quad (M1)$$

$$x = 0 \quad x = 4ap$$

$$y = p(4ap)$$

$$y = 4ap^2 \quad (1)$$

$$\text{iii) } m_{\text{tangent at Q}} = \frac{4ap}{2a} = 2p \quad (M1)$$

$$y - 4ap^2 = 2p(x - 4ap)$$

$$y - 4ap^2 = 2px - 8ap^2$$

$$y = 2px - 4ap^2 \quad (1)$$

$$\text{iv) Equation of tangent at P}$$

$$y = px - ap^2$$

$$px - ap^2 = 2px - 4ap^2 \quad (M1)$$

$$3ap^2 = px$$

$$3ap = x$$

$$y = p(3ap) - ap^2$$

$$y = 2ap^2 \quad (1) \quad R(3ap, 2ap^2)$$

$$(v) \quad x = 3ap$$

$$p = \frac{x}{3a}$$

$$y = 2a\left(\frac{x^2}{9a^2}\right)$$

$$y = \frac{2}{9a}x^2 \quad (1)$$

$$7(a) (i) \quad v = 6 - 2x$$

$$v^2 = 36 - 24x + 4x^2$$

$$\frac{v^2}{2} = 18 - 12x + 2x^2 \quad (M1)$$

$$\ddot{x} = -12 + 4x \quad \text{when } x=0 \quad \ddot{x} = -12 \quad (1)$$

$$(ii) \quad \frac{dt}{dx} = \frac{1}{6-2x}$$

$$t = -\frac{1}{2} \ln(6-2x) + C$$

$$0 = -\frac{1}{2} \ln 6 + C \quad C = \frac{1}{2} \ln 6 \quad (M1)$$

$$t = -\frac{1}{2} [\ln(6-2x) - \ln 6]$$

$$t = -\frac{1}{2} \ln \left(\frac{6-2x}{6} \right)$$

$$= -\frac{1}{2} \ln \left(1 - \frac{x}{3} \right) \quad (1)$$

$$(iii) \quad -2t = \ln \left(1 - \frac{x}{3} \right)$$

$$1 - \frac{x}{3} = e^{-2t} \quad (M1)$$

$$\frac{x}{3} = 1 - e^{-2t}$$

$$x = 3(1 - e^{-2t}) \quad (1)$$

$$(b) (i) \quad y = -gt + V \sin \alpha$$

$$t=0 \quad y = V \sin \alpha$$

At max height, $y=0$, $y=3h$

$$0 = -gt + V \sin \alpha$$

$$t = \frac{V \sin \alpha}{g} \quad (M1)$$

$$y = -\frac{g}{2} \left(\frac{V \sin \alpha}{g} \right)^2 + V \sin \alpha \left(\frac{V \sin \alpha}{g} \right)$$

$$3h = -\frac{V^2 \sin^2 \alpha}{2g} + \frac{(2)V^2 \sin^2 \alpha}{(2)g} \quad (M1)$$

$$6gh = V^2 \sin^2 \alpha$$

$$\therefore V \sin \alpha = \sqrt{6gh}$$

$$(ii) \quad \text{Let } y=0 \quad 0 = t \left(-\frac{g}{2} + V \sin \alpha \right)$$

$$t \neq 0 \quad t = \frac{2V \sin \alpha}{g} \quad (M1)$$

$$d = V \cos \alpha \left(\frac{2V \sin \alpha}{g} \right)$$

$$V \cos \alpha = \frac{dg}{2\sqrt{6gh}} \quad (1)$$

$$(iii) \quad x = V \cos \alpha \cdot t$$

$$t = \frac{x}{V \cos \alpha}$$

$$= \frac{2\sqrt{6gh} x}{g\sqrt{6gh}} \quad (M1)$$

$$y = -\frac{g}{2} \left(\frac{2x\sqrt{6gh}}{g\sqrt{6gh}} \right)^2 + \sqrt{6gh} \left(\frac{2\sqrt{6gh} x}{g\sqrt{6gh}} \right)$$

$$= -\frac{12hx^2}{d^2} + \frac{12hx}{d} \quad (M1)$$

$$= \frac{12hx}{d} \left(1 - \frac{x}{d} \right) \quad (M1)$$